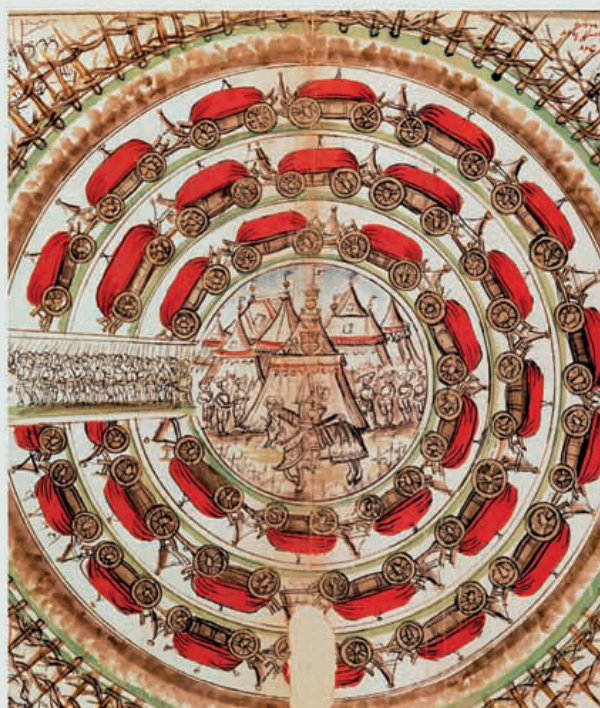


Years War. A revolution in fortifications, however, soon tamed the siege guns. By the 16th century, medieval castles with their tall curtain walls were being replaced by lower-lying thick-walled fortifications, with angled bastions (see pp.176–77) that provided the defenders with gun platforms to repel attack, enabling them to rake an enemy attempting an assault. Even after an additional use was found for gunpowder – packing it into a tunnel dug beneath wall, then exploding it to make a breach – besieging forces often found they could not regain the upper hand.

THE USE OF THE ARQUEBUS

A range of smaller, more mobile cannon was developed for use in field battles, along with a number of hand-held infantry and cavalry gunpowder weapons. During the 16th century, one such weapon – the arquebus, a firearm with a matchlock mechanism – equipped numbers of infantrymen in armies from western Europe to India and Japan. Slow-firing, inaccurate, and unreliable, arquebuses and the matchlock muskets that eventually succeeded them by no means dominated the battlefield, and instead coexisted with pikes, edged weapons, and crossbows. Nor did the increased use of hand-held guns and cannon make the medieval armoured knight instantly redundant. Charges by lance-wielding knights were still being



◀ THE HUSSITE WAGENBURGEN

Forerunners of modern armoured vehicles, these heavy carts were used to provide protection and mobility for the Hussite soldiers, who rode in them to battle armed with simple gunpowder weapons.

executed well into the late 16th century. Instead, the adoption of firearms was a gradual process. It was only from the 1540s onwards that wheellock pistols started to become the standard equipment of the cavalryman. And it would not be until the late 17th century that all infantrymen were equipped with firearms.

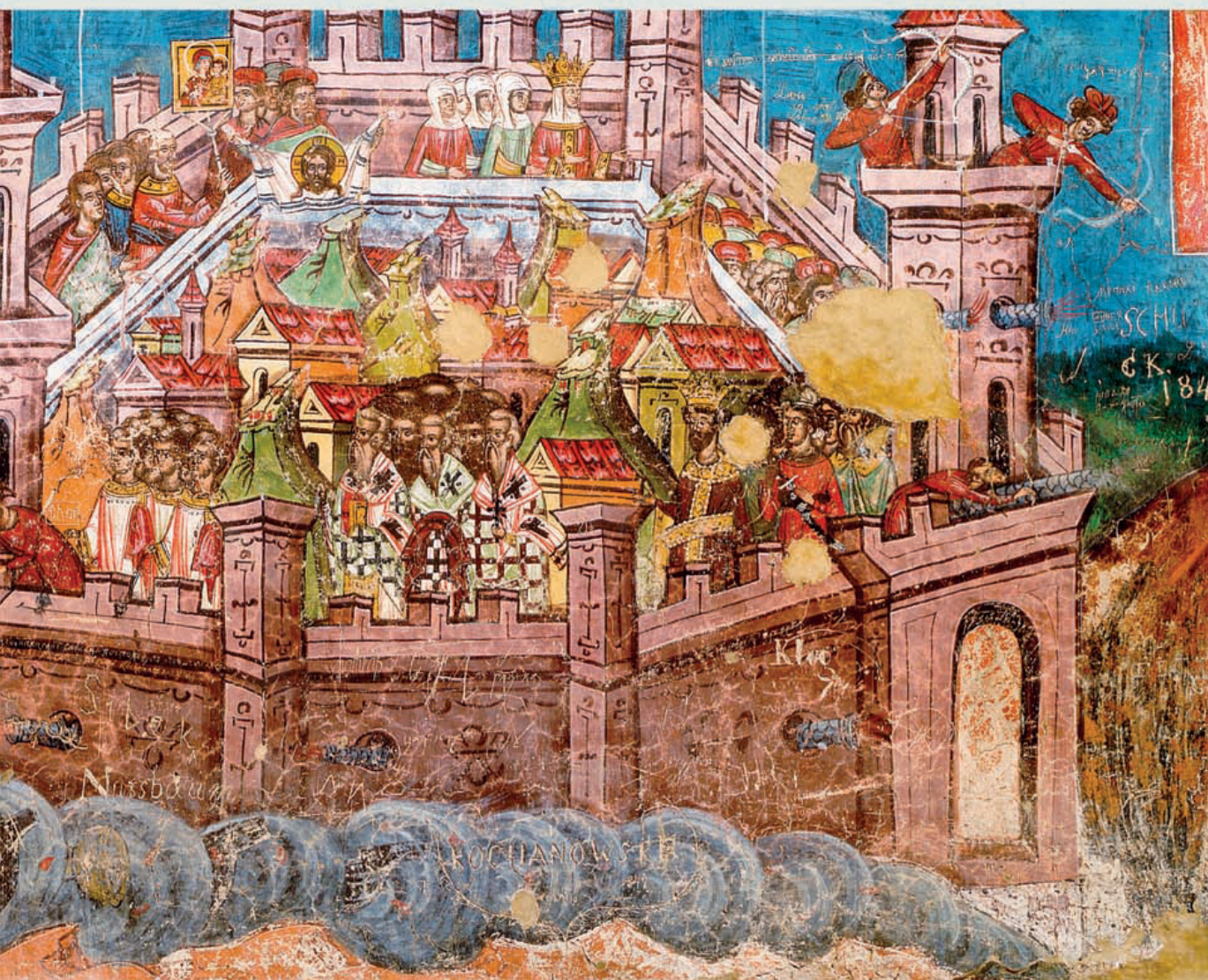
KEY FIGURE

ODA NOBUNAGA 1534–82

Against the backdrop of 16th-century feudal Japan, hard-headed *daimyo* (warlord) Oda Nobunaga broke from the prevailing samurai tradition, which favoured the sword and the bow. Using volleys of arquebus fire at the Battle of Nagashino, in 1575, he defeated the rival Takeda clan, demonstrating that gunpowder could be a significant force in battle.



▲ Oda Nobunaga introduced rotating volleys of fire together with the wooden stockade for defence.



◀ THE FALL OF CONSTANTINOPLE

The Ottomans' powerful cannon played a large role in the capture of Constantinople and the subsequent fall of the Byzantine Empire.